

Limitations Analysis

One major limitation of using a site like Grad Café is that it relies on people to voluntarily report. Because it is voluntary, and there is no method to verify information, there is a high chance of selection bias within the data. Applicants who are more likely to submit a report might be individuals with really strong credentials, admitted to a prestigious program or engaged enough in the applicant community to share their outcomes are more likely to post. This can explain why from the data that was scraped the GRE Quantitative score is very high with an average of 165 with the max being 170. The platform does not contain a random group of all applicants but a group that chose to report themselves with many being high-performers and selective institutions. This also results in any averages that are being calculated such as GPA or GRE not truly indicative of all applicants' performance.

Another limitation is that due to submissions relying on voluntary reports there are data quality issues. From the overall data during initial scrape, there has been a lot of information entered inconsistently such as university names either being abbreviations such as JHU or some entries spelling out the university name such as Johns Hopkins or Johns Hopkins University. Missing data is another issue such as GPAs and GREs were absent from many entries and this meant the results for Question 3 do not accurately reflect the slice of entries that were scraped but only a small portion of the scraped portion of applications.